

AI/ML/DL Trends & Portfolio Plan 2026

Full-stack AI SaaS Founder | Low-cost SLM/Edge Stack

Field	Value
Date	2026-06-12
Version	1.0
Target Role	Full-stack AI SaaS Founder
Constraint	Low-cost (SLM / edge / open-source)
Sources	30+
Confidence	High

Research: deep-research + brainstorming-research-ideas skills

Executive Summary

• Paradigm shift 2026: dari demo chatbot ke production agent harness.

• Tren mainstream: multi-agent systems, DSLMs, SLM+edge, trajectory eval, MCP/A2A, physical AI.

• Niche diferensiasi: hybrid SLM router, agent harness, confidence scoring, GEO.

• Job market: recruiter cari end-to-end production (deploy, monitor, eval, cost).

• 3 flagship projects: DocFlow, AgentOps Lite, LogSentry AI.

• Timeline: 12-16 minggu hingga 2-3 live SaaS dengan real users.

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1. Metodologi Riset

Framework dari skills di GLOBAL SKILLS AI AGENTS/RESEARCH: deep-research (multi-source synthesis), brainstorming-research-ideas (tension hunting, What Changed?), research router (classification + fallback).

Skill	Fungsi
deep-research	Multi-source synthesis, citations
brainstorming-research-ideas	Tension hunting, niche discovery
research router	Classification + fallback workflow

Sumber primer: Gartner, Forrester, Deloitte 2026; job templates KORE1, BirJob; arXiv agent papers; SLM production guides.

2. Tren HOT 2026 (Mainstream)

Paradigm Shift

DEMO CHATBOT --> PRODUCTION AGENT HARNESS (memory + guardrails + eval + deploy)

Tren Utama

#	Tren	Kenapa Hot
1	Multi-Agent Systems	Workflow bisnis kompleks via agent spesialis
2	Agent Harness	Runtime: auth, memory, budget, HITL, observability
3	DSLMS	Akurasi + compliance per industri
4	SLM + Edge AI	10-30x lebih murah, latency <50ms
5	Trajectory Evals	Eval tool-call path, bukan final answer saja
6	MCP + A2A	Standard tool integration antar agent
7	Physical AI	Robot, drone, smart equipment
8	AI Security Platforms	Visibility + control untuk AI apps
9	Hybrid Inference	SLM 90% lokal, cloud 10% edge case
10	AI-Native Dev	Tim kecil build software dengan GenAI

Sinyal Adjacent

Synthetic data, neuromorphic computing, GEO, AI wearables, confidential computing, continual learning, world models.

3. Niche & Underdiscussed Topics

Niche	Tension	Peluang Portfolio
Agent Harness Eng.	Reliability vs autonomy	Circuit breaker, budget cap
Trajectory Eval	Quality vs deploy speed	Golden dataset dari traces
Hybrid SLM Router	Cost vs capability	Qwen3-4B lokal, escalate 10%
Mech. Interpretability	Trust vs black-box	Uncertainty probe classifier
World Models	Data scarcity vs quality	Synthetic scenario generator
Continual Learning	Static vs evolving	User correction -> LoRA batch
Digital Provenance	AI content vs trust	Audit log per action
DeepConf Scoring	Know when unsure	Auto-escalate ke human
GEO	SEO vs AI search	Brand visibility audit tool
Confidential Computing	Privacy vs cloud	On-device + encrypted embed

Rekomendasi Triple Niche

• Hybrid SLM Router

• Agent Harness

• Trajectory Eval

4. Job Requirements ML/AI/LLM Engineer 2026

Hard Skills - Semua Role

- Python (typing, async)
- Docker + CI/CD
- SQL + pipelines
- Cloud (AWS/GCP/Azure)
- Monitoring drift/latency/cost

ML Engineer

- PyTorch/TF end-to-end
- Model serving (Triton)
- MLOps (MLflow, W&B)
- Classical ML (XGBoost)
- LoRA fine-tuning

LLM Engineer

- RAG + pgvector
- LangGraph agents
- RAGAS + trajectory eval
- Langfuse/OTel
- Cost routing

Full-stack AI SaaS

- Supabase Auth + RBAC
- Stripe billing + metering
- Product metrics
- Domain knowledge

Soft Signals

- Live demo + real users
- Documented trade-offs
- Eval numbers published
- Architecture diagram
- Open-source harness component

Role	US Total Comp 2026
LLM Engineer	\$170K - \$420K

Role	US Total Comp 2026
ML Engineer	\$160K - \$380K
AI Engineer	~\$149K

5. Portfolio Projects

Kriteria Wajib (Bukan Demo)

- Auth + free/paid tier
- Public URL deployed
- Real user acquisition
- Monitoring + error tracking
- Eval + cost docs

Tier 1 - Build First

DocFlow - Hybrid RAG UMKM

OCR -> RAG -> hybrid router 90% lokal. Target: accountant, UMKM. Cost: \$10-25/bulan.

AgentOps Lite - Harness + Eval SaaS

SDK harness, trace viewer, CI eval gate. Target: indie devs. Cost: \$15-30/bulan.

LogSentry AI - Hybrid Log Classifier

XGBoost 95% + SLM explain + cloud P1. Target: solo devops. Cost: \$10-20/bulan.

Tier 2

- PitchMind - GEO content audit
- VoiceBridge ID - voice agent Indonesia
- ModelRegistry Hub - MLOps solo engineers

Tier 3

- ContinualLearn Feed - batch LoRA adaptation
- SynthForge - synthetic data generator

6. Roadmap Implementasi (12-16 minggu)

Minggu	Milestone
1-2	Monorepo, Supabase, Stripe, Ollama, CI/CD
3-6	DocFlow MVP + 5 beta users
7-10	AgentOps Lite + Product Hunt launch
11-14	LogSentry AI
15-16	Case studies + demo video

7. Architecture Template

Next.js App
|
FastAPI Gateway --> Supabase Auth + Stripe
|
Hybrid SLM Router
|-- Ollama Qwen3-4B (90%)
|-- Cloud LLM API (10%)
|
Agent Harness --> PostgreSQL + pgvector
|
Langfuse + Sentry + Prometheus

8. Perbandingan vs Tutorial Biasa

Tutorial	Portfolio Anda
Jupyter notebook	Deployed SaaS + billing
Chat with PDF demo	Hybrid router + cost metrics
LangChain hello world	Harness + circuit breaker
Akurasi tanpa konteks	Trajectory eval + replay
GPT-4 only	SLM-first documented
No users	5-20 beta users

Appendix A: Sources

1. Gartner Top Strategic Technology Trends 2026
2. Forrester Top 10 Emerging Technologies 2026
3. Deloitte Tech Trends 2026
4. Agentic Lexicon Jan-Feb 2026
5. VoltAgent Awesome AI Agent Papers 2026
6. Zylos SLM Production & Edge AI 2026
7. Dell Edge AI Predictions 2026
8. CODERCOPS SLM Edge Deployment
9. LLM Observability & Evals 2026
10. HarnessAudit (UCSB)
11. Production Agent Harness (DEV Community)
12. Agent Harness Observability (Rahul Kashyap)
13. Mechanistic Interpretability 2026 (Towards AI)
14. AgenticCareers LLM Engineer Guide 2026
15. Howdy AI vs ML Engineer 2026
16. KORE1 ML Engineer JD 2026
17. BirJob ML Engineer Roadmap 2026
18. Codebasics Production AI Projects 2026
19. InfraSketch ML System Design Patterns
20. GigaWorld-0 World Models (arXiv)
21. ARROW Continual RL (arXiv)

Appendix B: Tech Stack

Category	Primary	Alternative
SLM Runtime	Ollama	llama.cpp
SLM Models	Qwen3-4B, Phi-4-mini	Gemma 3
Vector DB	pgvector	ChromaDB
Agents	LangGraph	CrewAI
Eval	RAGAS + custom	Braintrust
Tracing	Langfuse	LangSmith
Auth	Supabase	Clerk
Billing	Stripe	LemonSqueezy
Deploy	Railway	Fly.io

Appendix C: Cost Estimation

Project	Infra/mo	API/mo	Total/mo
DocFlow	\$10-15	\$0-5	\$10-25
AgentOps Lite	\$15-20	\$0	\$15-30
LogSentry AI	\$10-15	\$0-5	\$10-20
PitchMind	\$10	\$10-20	\$20-30
VoiceBridge ID	\$20-30	\$10-30	\$30-60

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